Joshua Gaus

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Objective

Electrical engineering undergraduate with hardware prototyping and applied machine learning experience. Proficient in pipeline development from diverse data streams. Eagle Scout with ample project management, public speaking, and leadership experience. Seeking an internship for Summer 2027 in hardware engineering and machine learning.

Education

Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*
Bachelor of Science in Electrical Engineering Expected Graduation: May 2030

Blessed Trinity Catholic High School | Roswell, GA *August 2022 – May 2026*
Valedictorian (1st of 229)
GPA 4.59

Projects

Lithium-Ion Battery Remaining Useful Life (RUL) Prognostics & Telemetry Node

- Engineered a predictive machine learning pipeline using Random Forest regression and a SHAP analysis on NASA battery aging datasets to forecast Li-ion cell RUL from a 5-minute discharge window.
- Designed a battery discharge characteristic telemetry node using an STM32 Nucleo-64 microcontroller, INA219 current/voltage sensor, and C firmware to capture real-time cell performance under passive resistive loads.

Satellite Land-Use Image Classification

- Designed an end-to-end deep learning pipeline to classify 27,000+ satellite imagery samples (EuroSAT dataset) into distinct land-use categories.
- Trained a Convolutional Neural Network (CNN) and optimized layer architecture, learning rates, and regularization.

Technical Skills

Programming Languages: Python, Java, C, HTML/CSS
Hardware: STM32 microcontrollers
Licenses: FCC Technician Class HAM license

Experience

Hen Mother Cookhouse | Alpharetta, GA *December 2025 – May 2026*
Table Busser

- Demonstrated reliability, professionalism, and punctuality while working 16 hours per weekend as a full-time student.
- Proactively sought feedback from managers, enhancing productivity and improving collaboration with coworkers.

Blessed Trinity Student Tutor Club | Roswell, GA *August 2025 – May 2026*
Head Tutor

- Implemented a tutor signup and tracking system that organized over 200 tutoring sessions per year.
- Improved communication and critical thinking skills by making difficult topics in chemistry more accessible to students.

Activities

Scouting America | Eagle Scout *May 2019 – May 2024*

- Benefited over 50 seniors with dementia and children with special needs by leading 15 volunteers in constructing 24 custom-made physical therapy lapboards and six shelving units.
- Mentored younger scouts in developing resourcefulness, self-reliance, and problem-solving skills during 50+ days of camping, backpacking, and hiking.

Theta Xi Fraternity | Freshman Social Chairman *August 2026 – Present*

- Coordinate and organize weekly multi-organization events for a 30-person member class, managing logistics, preparation, and activity execution.
- Boost member engagement by pitching and delivering innovative programming (e.g., structured dance lessons, interactive performance competitions) resulting in peak turnouts of 60+ attendees.